

From Data to Dissemination: Creating Dashboards for Buffalo African-American Community-based Participatory Air Monitoring

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1. From Numbers to Information

Fine particulate matter (PM_{2.5}) poses significant health threats, and its spatial and temporal variability can be substantial. On the East Side of Buffalo, New York, the predominantly African-American residents face a disproportionate burden of chronic disease, which may be worsened by poor air quality. In the absence of localized monitoring, however, neighborhood-level pollution exposures have remained largely ignored. A small number of EPA sensors serves the entire Buffalo-Niagara metropolitan area, obscuring the spatio-temporal variability of air pollution and preventing vulnerable community members access to accurate air quality information.

To address this gap in data and equity, we partnered with local community members using community-based participatory research (CBPR) methods. We deployed a network of low-cost PurpleAir(PA) sensors across the East Side, generating high-resolution PM_{2.5} data. Recognizing that raw data alone is insufficient, we co-designed an interactive dashboard with community members to visualize pollution patterns in real time, making air quality insights accessible, understandable, and relevant to those most affected.

2. Objectives

To design an effective interface for spatio-temporal variability of air quality for disadvantaged community members

- Co-design with community members to accommodate their mental models and priority needs for visualizing spatio-temporal air quality patterns
- Develop an interactive dashboard prototype that reflects those priorities and is easy to navigate

3. Methods

Data Collection Strategy

- Deployed 30 PA sensors in the study area
- Built a fully automated data collection pipeline
- Stored the data in a hosted Database, which can be retrieved by queries at anytime

Dashboard

- Refined the interface through community-led sessions to capture what users need
- Incorporated accessible design with users in mind
- Used GitHub, ensuring easy deployment and reproducibility

Figure 2: Architecture Overview

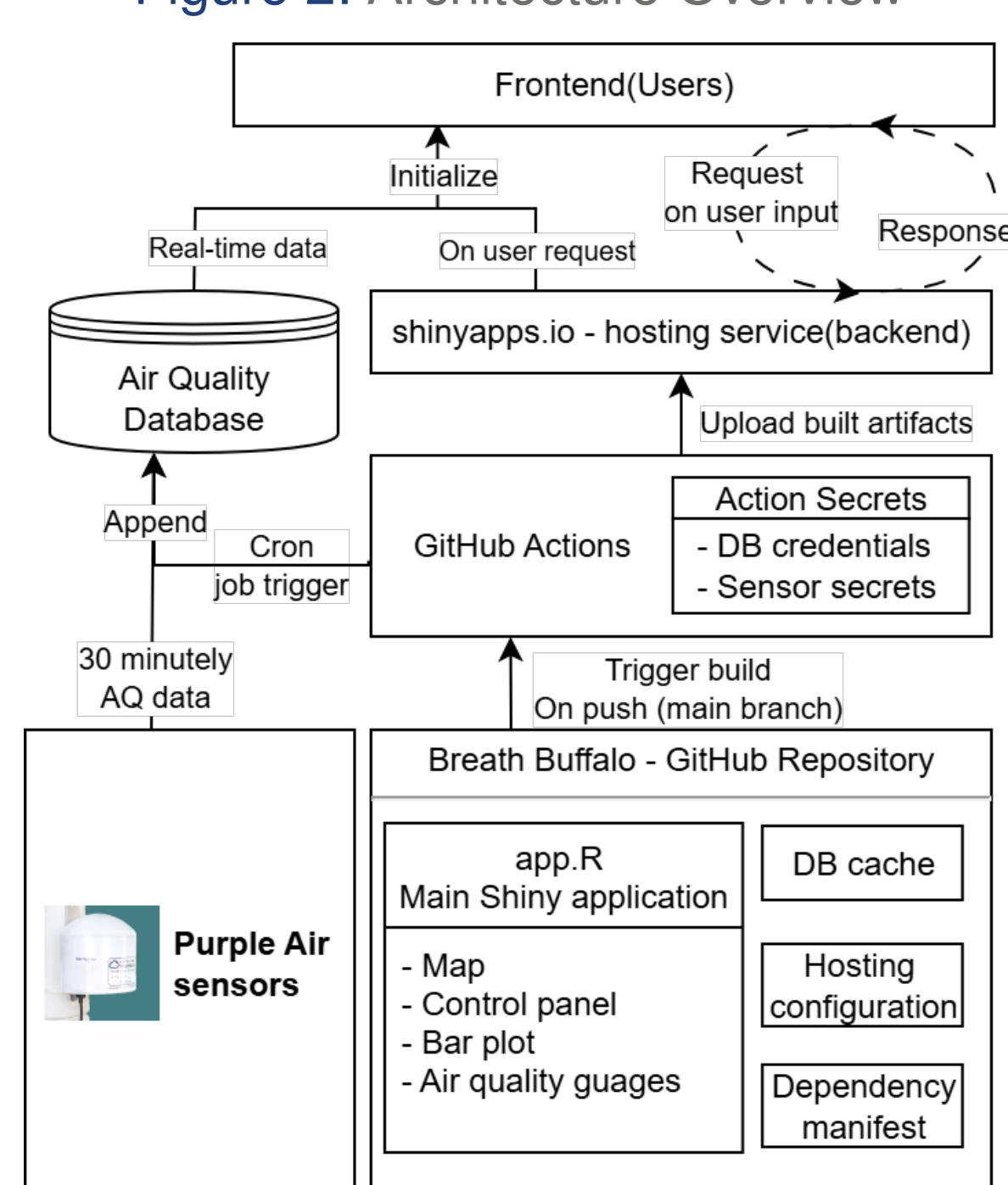


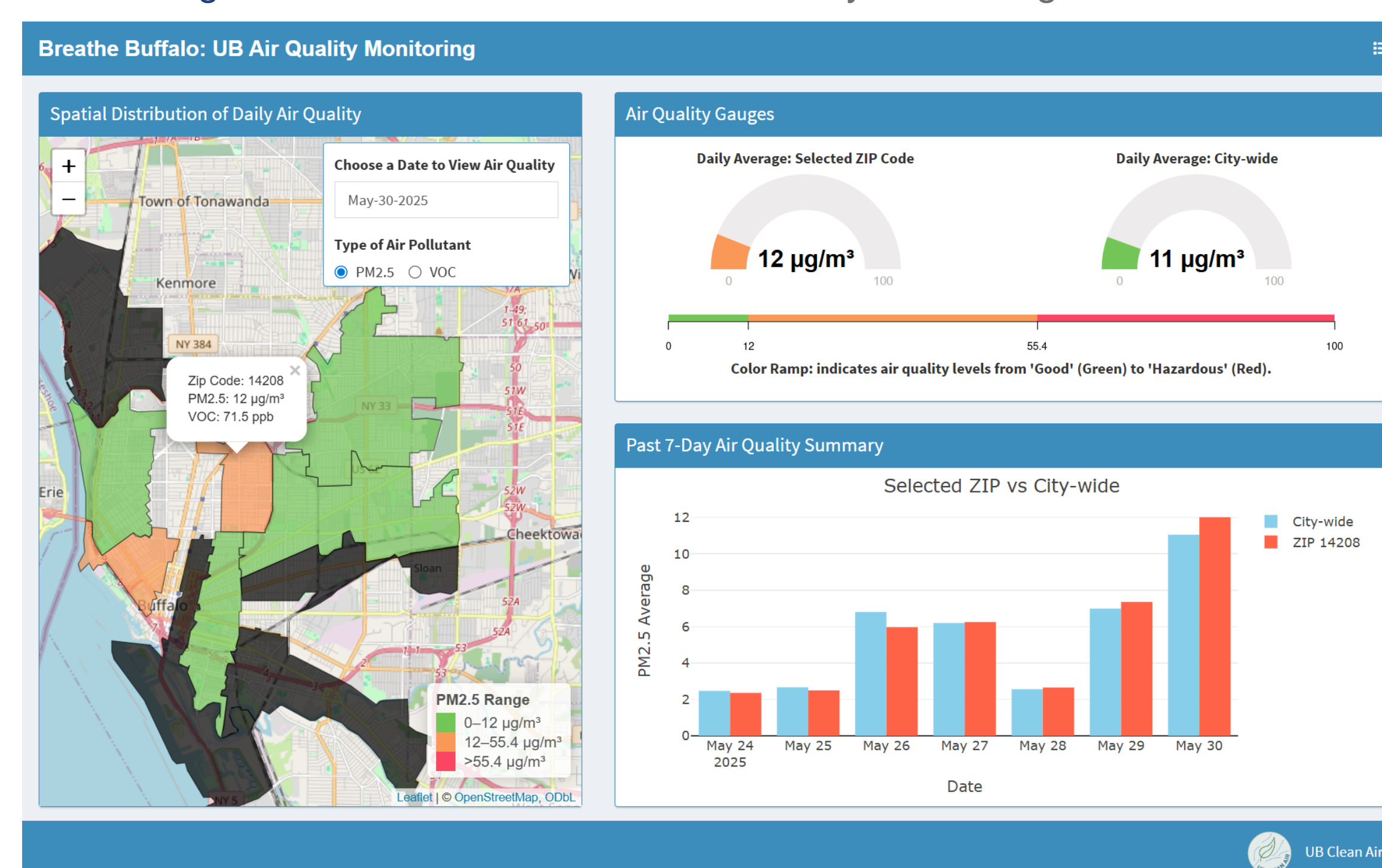
Figure 4: Calibrating PA Sensors



Figure 3: PurpleAir Sensor



Figure 1: Breathe Buffalo: UB Air Quality Monitoring Dashboard



4. Results

Visualizing Air Quality

- **Visualization of Localized and Dynamic Variability of PM_{2.5}:**
Shows the daily and community specific air quality using a map and graphics
- **Comparison With The Baseline:**
Provides both the community-specific air quality information and the city-wide air quality from EPA monitoring sites

User-friendly Dashboard

- **Color-coded Components:**
Displays air quality data using a user-friendly color scheme in a map, a bar plot, and gauges with a reference color ramp
- **Interactive Dash:**
An interactive map with pop-up information and the controller for selecting the time and location

5. Discussion

- Co-developed a community-informed dashboard for real-time, accessible air quality data.
- Empowered residents by providing localized and dynamic information on air pollution.
- Future works can be expanded by using AI:
 - AI voiceover with multi-language support
 - Estimate air quality in areas without measurements using machine learning algorithms

References

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- Yan, D., Ji, H., Fu, H., Jiang, J., Su, B., & Ye, B. (2024). The effect of fine particulate matter (PM_{2.5}) pollution on health inequality: an intergenerational perspective. *Environmental Geochemistry and Health*, 46(6), 195.

Give it a try yourself!

Breath Buffalo



Source Code (GitHub)

